

OpineSum: Entailment-based self-training for abstractive opinion summarization

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Introduction

01

- **OpineSum**

- ✓ Abstractive Opinion summarization : **Entailment-base self-training**

- ❖ Textual entailment and capture the consensus of opinions

- ❖ Train both unsupervised and few-shot learning

Related Work

03

- Unsupervised neural networks
 - ❖ Auto-encoder : input data **Encoder** Focus
 - ❖ VAEs(Variation Auto Encoders) : Generate **Decoder**
- Self-training methods
 - Unlabeled data features, auto labeling, pretraining and fine-tuning
- Few-shot learning : k-way n-shot , Similarity learning

Textual entailment to identify consensus

05

- Defining review consensus
 - ✓ find out how often the collection supports the proposition,
- Extracting propositions
 - ✓ split review sentences into propositions and use our key units
- Scoring consensus
 - ✓ $S(m_j) = \sum_{1 \leq i \leq N} E(R_i, m_j)$, review R_i , proposition m_j
- Silver summaries
 - ✓ MMR(Maximal Marginal Relevance)

Consensus opinion

01

- **Consensus or common opinion**

✓ the SPACE corpus (on TripAdvisor.com)

“location is right across balboa park”

: summary-worthy one

* create our silver-standard data

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R1	...very large and clean with a nice size kitchen. The hotel is located right across the street from balboa park and within walking distance of a rite aid drug-store..
R2	...If you insist on staying here, reserve a refurbished room and get that promise in writing! The location was great for tourists, right across from balboa park. You could walk to the zoo (about 1/4 mi)...
R3	...I decided to stay at the park manor suites hotel since it seemed to be close to san diego zoo. The hotel is conveniently located in front of balboa park, walking distance to san diego zoo,...
R4	...The staff are both pleasant and professional. Hotel is across from balboa park on sixth ave. This is the park west area, and features a diverse array of restaurants...
R5	...As other reviewers have said, it's very easy to be here without a car - balboa park is just across the road and the airport is a short taxi ride away.

Consensus opinion

01

- **Consensus or common opinion**

- ✓ Same claim may be expressed

“This hotel is in the heart of Times Square ” ,
‘Hotel’s location is slap bang in the middle of Times Square.’

“The fish is tasty” and ‘The salmon is delicious’

Consensus opinion

01

- **Extracting propositions**

- ✓ Long review sentences : contain a bunch of different claims

- > split sentences into propositions : as a 'single claim or fact'

- (* conjunctions, period, and comma subject to a length of four)

Review sentence	Extracted propositions
There was loads of cupboard space and a fantastic easy to use safe.	There was loads of cupboard space and ₁ a fantastic easy to use safe. ₂

Consensus opinion

01

- **Scoring consensus**

- ✓ the M propositions

- **textual entailment relation as $P \rightarrow H$** (P is a premise and H is a hypothesis)

$$S(m_j) = \sum_{1 \leq i \leq N} E(\underset{\text{review}}{R_i}, \overset{\text{proposition}}{m_j})$$

- ✓ BERT-large : [CLS] Token (a linear layer to predict three classes: entailment, contradiction and neutral)

- ✓ Apache Beam1 pipeline

Consensus opinion

01

- **Silver summaries**

- decreasing order of their scores $S(mi)$
- take the top n as the silver summary sentences
- edundancy removal technique : MMR (maximal marginal relevance)

$$overlap(p_k, s_i) < 2, \quad \forall i, 1 \leq i \leq |S|$$

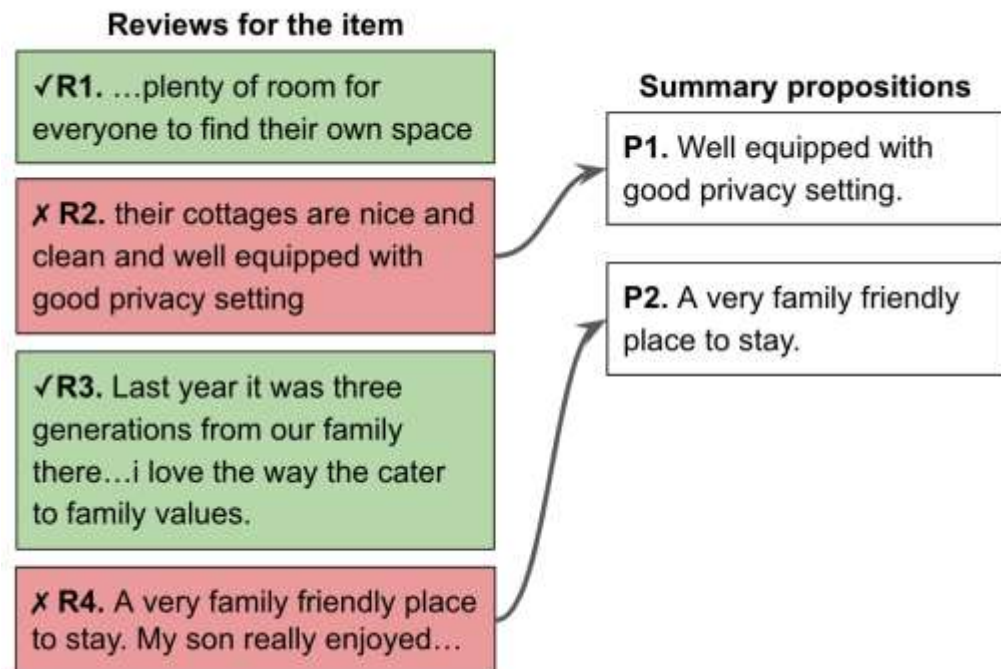
p_k Highest scoring proposition

Consensus opinion

01

- **Silver summaries**

- P1 was extracted from R2
- P2 from R4



Q & A

Experiments

05

- Dataset
 - ❖ Unlabelled review corpus
 - ❖ Evaluation dataset containing with human summaries
 - ✓ SPACE-unlabeled : 1.1 million reviews for 11,000 hotels
 - ✓ SPACE-eval : human generated summaries for a smaller set of 50 hotels
- Models
 - ❖ T5 (**Text-to-Text Transfer Transformer**)
 - ✓ Pre-trained encoder-decoder models based on T5's framework
 - ✓ LongT5 : polynomial scale relationship to input length

Experiments

05

- Evaluation
 - ✓ Silver Data (as **SPACE-OpineSum**) : the unlabelled review corpus
 - SPACE items with a minimum of 50 reviews (4,729 items)
 - Computed 1.3B entailment predictions
 - ✓ Self-training
 - Unsupervised and few-shot learning abstractive summarization
 - Trained **LongT5-Large** (770M parameters) and LongT5-(Large, XL)
 - Compare these systems with prior unsupervised work in the **SPACE-eval dataset**
 - the ROUGE performance on the validation set

Experiments

05

- Evaluation
 - ✓ Few-shot Learning
 - SPACE-eval (25 total) : training set (15items) and validation set(10 items)
 - Reduced learning rate, 1/5th of the standard $1e-4$

Results

06

OPINESUM system use self-training only

Model	R1	R2	RL
Previous systems			
Lexrank	36.86	8.81	22.96
Acesum	42.64	14.50	25.20
OPINESUM systems			
LongT5 Large	45.84	16.30	29.18
LongT5 XL	43.41	13.82	23.84

Table 5: Results for the unsupervised setting. The OPINESUM systems use self-training only and *no gold summaries*.

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Model	Checkpoint	R1	R2	RL
LongT5 L	Vanilla	45.51	13.03	29.28
LongT5 L	OPINESUM	47.19	14.60	30.13

Table 7: Results for the few-shot learning setting. All the models were finetuned on a small set of 15 training examples described in Section 6.3. ‘Vanilla’ systems are warm started from public checkpoints and do not see self-training data.

Results

06

ACESUM (Amplayo et al., 2021)

The staff were very friendly and helpful. the room was clean and clean. it was a great place to stay. if you want to stay in cicerone, it is a good place to get to the shopping area. there are many restaurants, restaurants and restaurants. but the staff are very friendly, friendly and friendly. they are a very nice hotel, a nice place to eat, and a lot of good food, as well as a small restaurant, the breakfast was very good, but a bit of.

previous best system (ACESUM)

OPINESUM-unsupervised

The hotel is located within walking distance of the Vatican. The rooms were clean and comfortable. The Cicerone is a nice hotel. As far as the hotel goes. The reception area is nice but the rooms. The breakfast buffet was fine. The room was a good size.

our unsupervised and few-shot self-trained

OPINESUM-fewshot

The staff was friendly and helpful. The rooms and hotel itself is modern, extremely clean! The rooms are a good size, with comfy beds, a breadth of amenities such as a great shower and a comfortable bed. The breakfast buffet is average, but very good, with lots of variety. The location is very central. The hotel is within walking distance of the Vatican and Piazza Navona. The Cicerone is a beautiful hotel, but the hallways need refurbishing.

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Conclusion

- simple self-training approach both unsupervised and few-shot
- using the **entailment weights** (scores) of each proposition
- self-training models could serve as **checkpoints**

Thank you!